



NextAIRE

Next Generation AI Researchers for Air Quality Excellence



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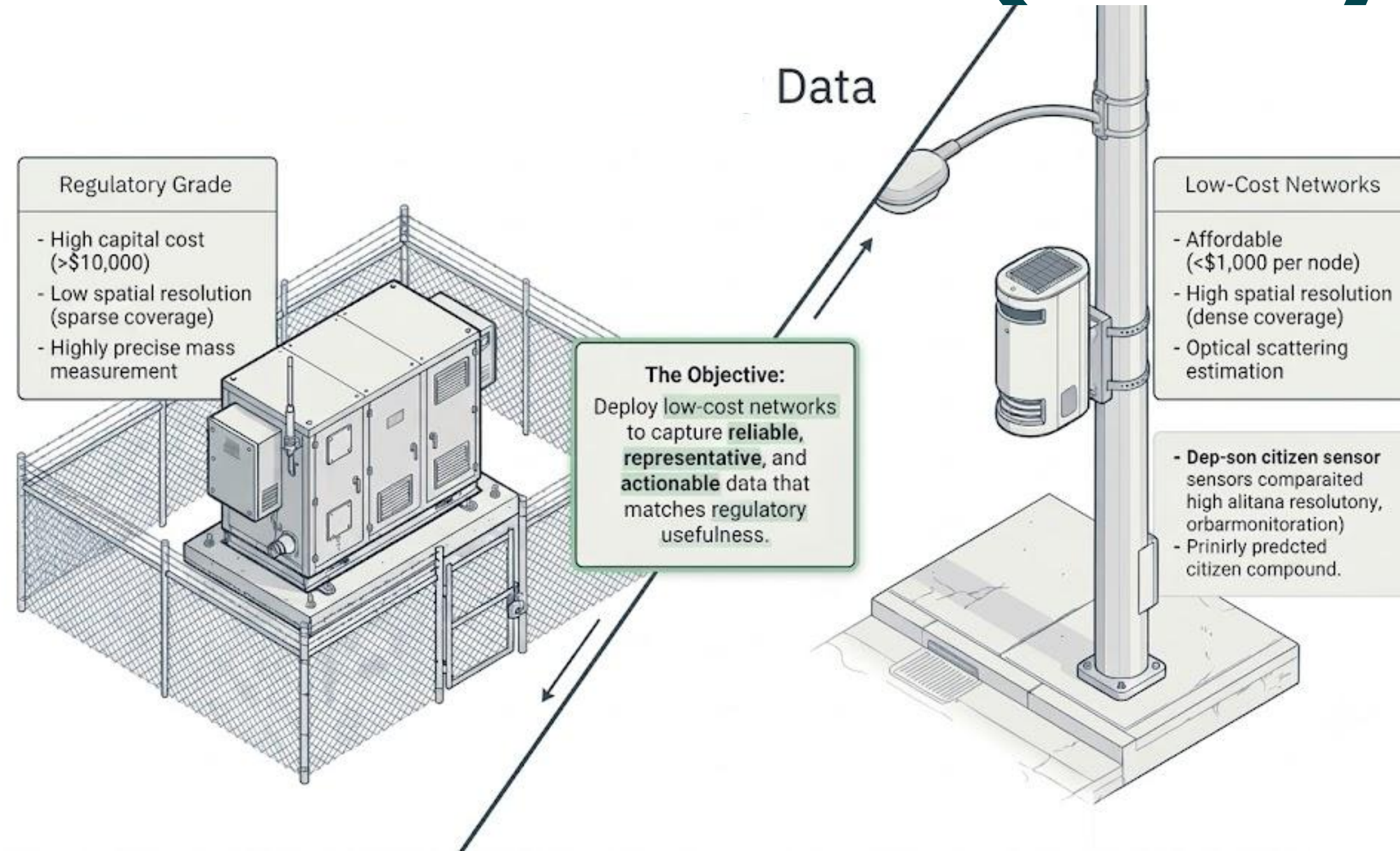
Next Generation AI Researchers for
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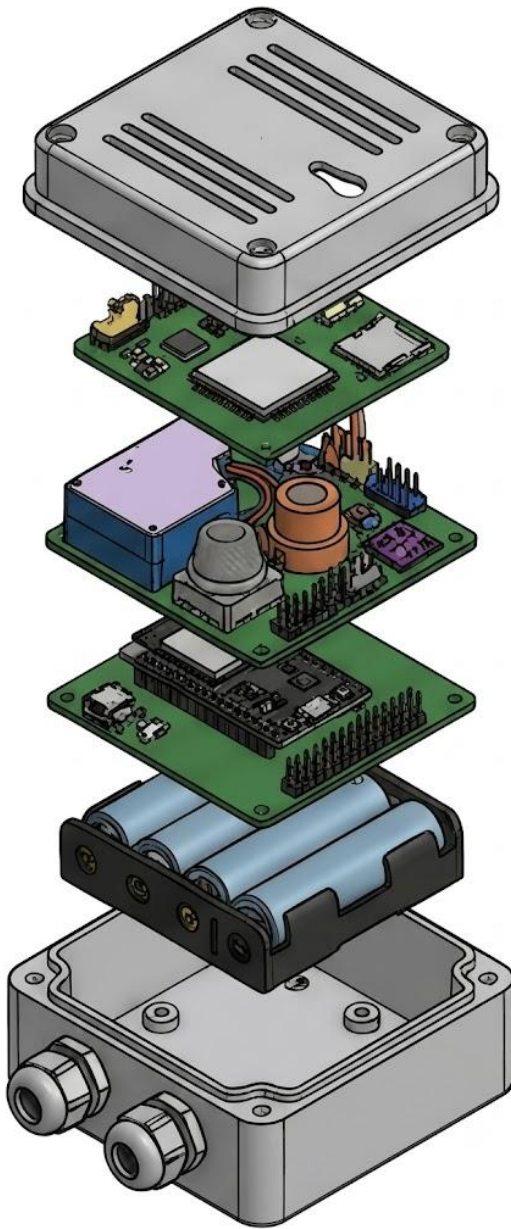
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Sensors – operations, siting and deployment

The democratization of Air Quality Data

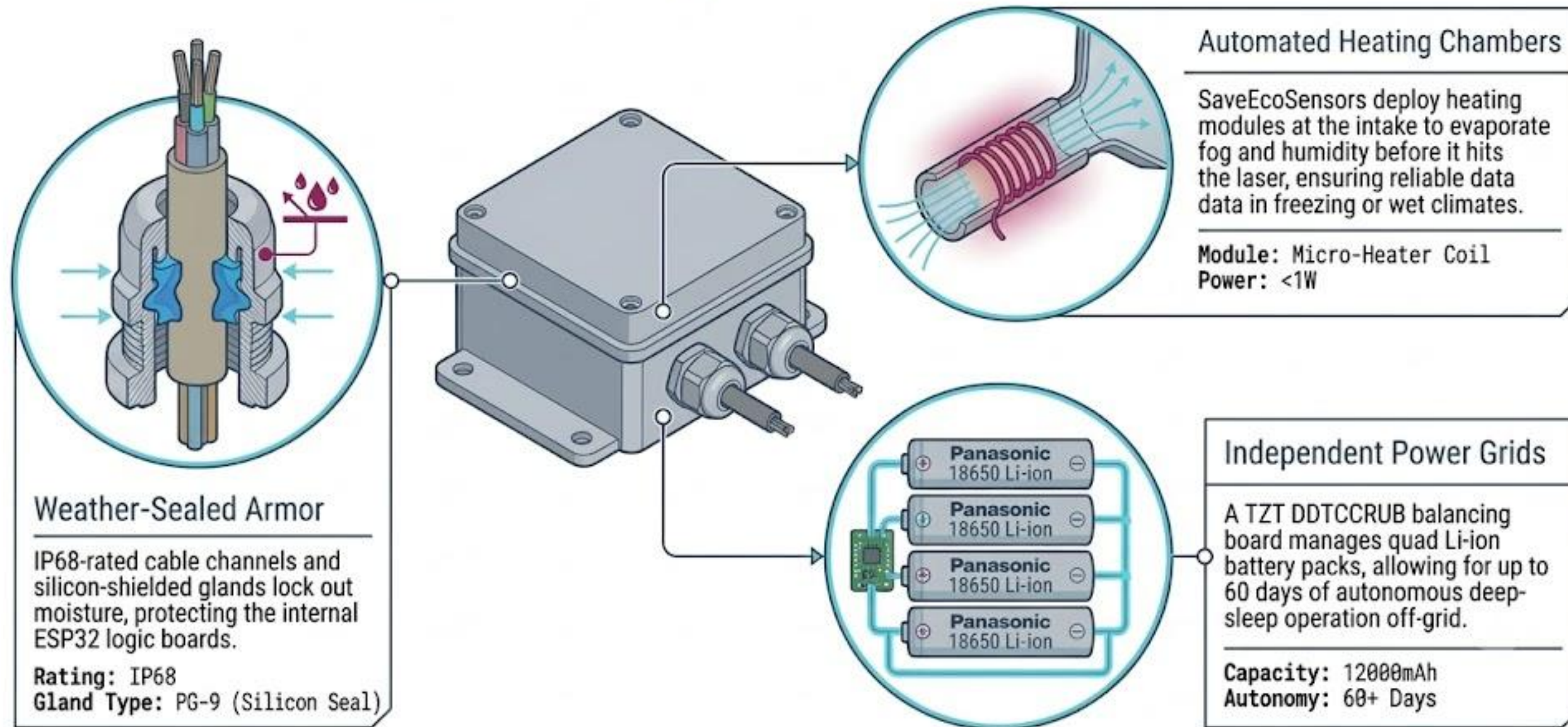


Anatomy of a Sensor

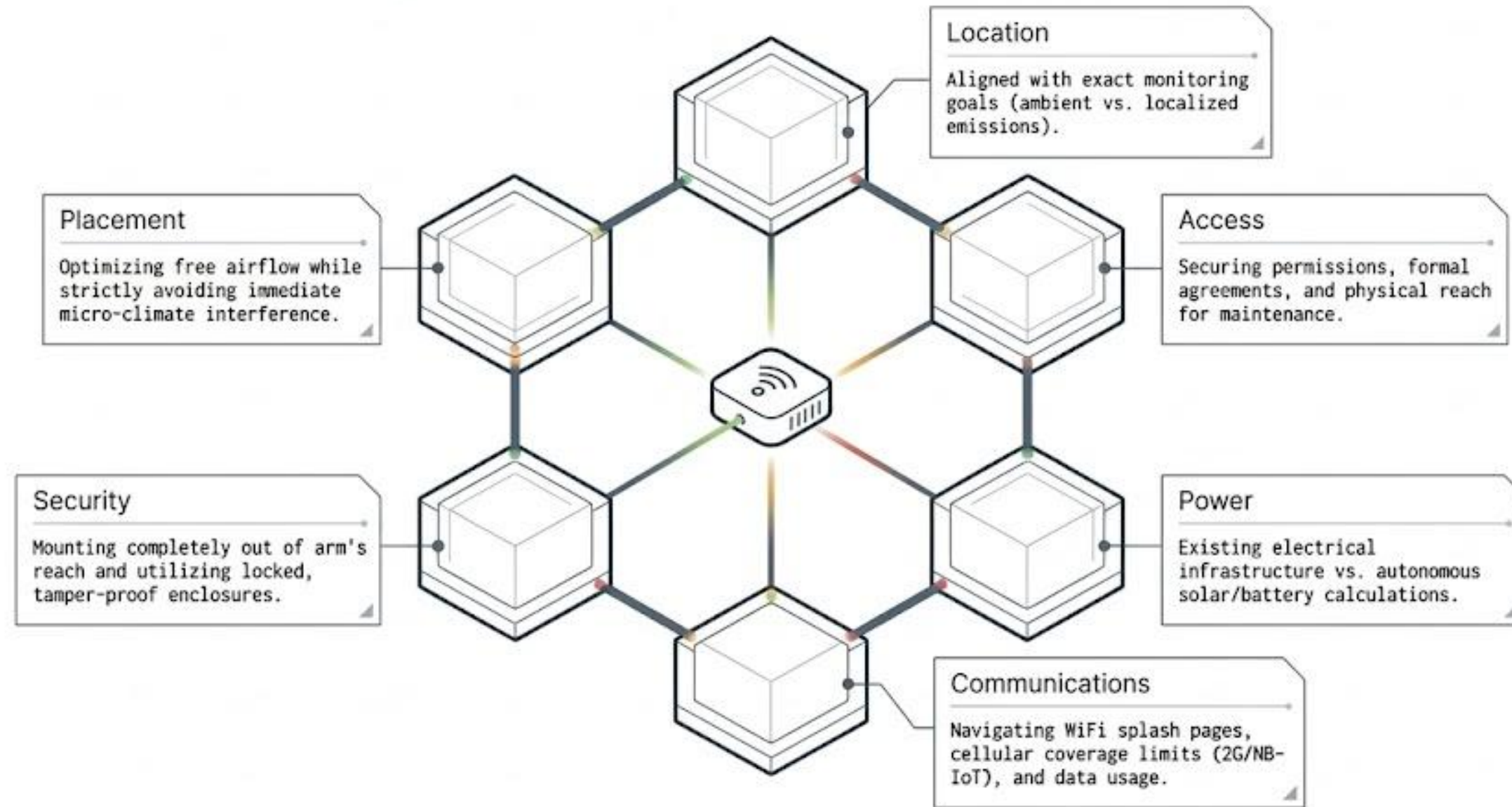


- Comms – MicroSD modul for offline storage, SIM780G fro IoT cellular
- Sensors – Gas sensors (CO, O3, NO, NO2)
- Brain – ESP32 DevKit v1
- Power – Li-on battery
- Housing – 3D printed or aluminum enclosure

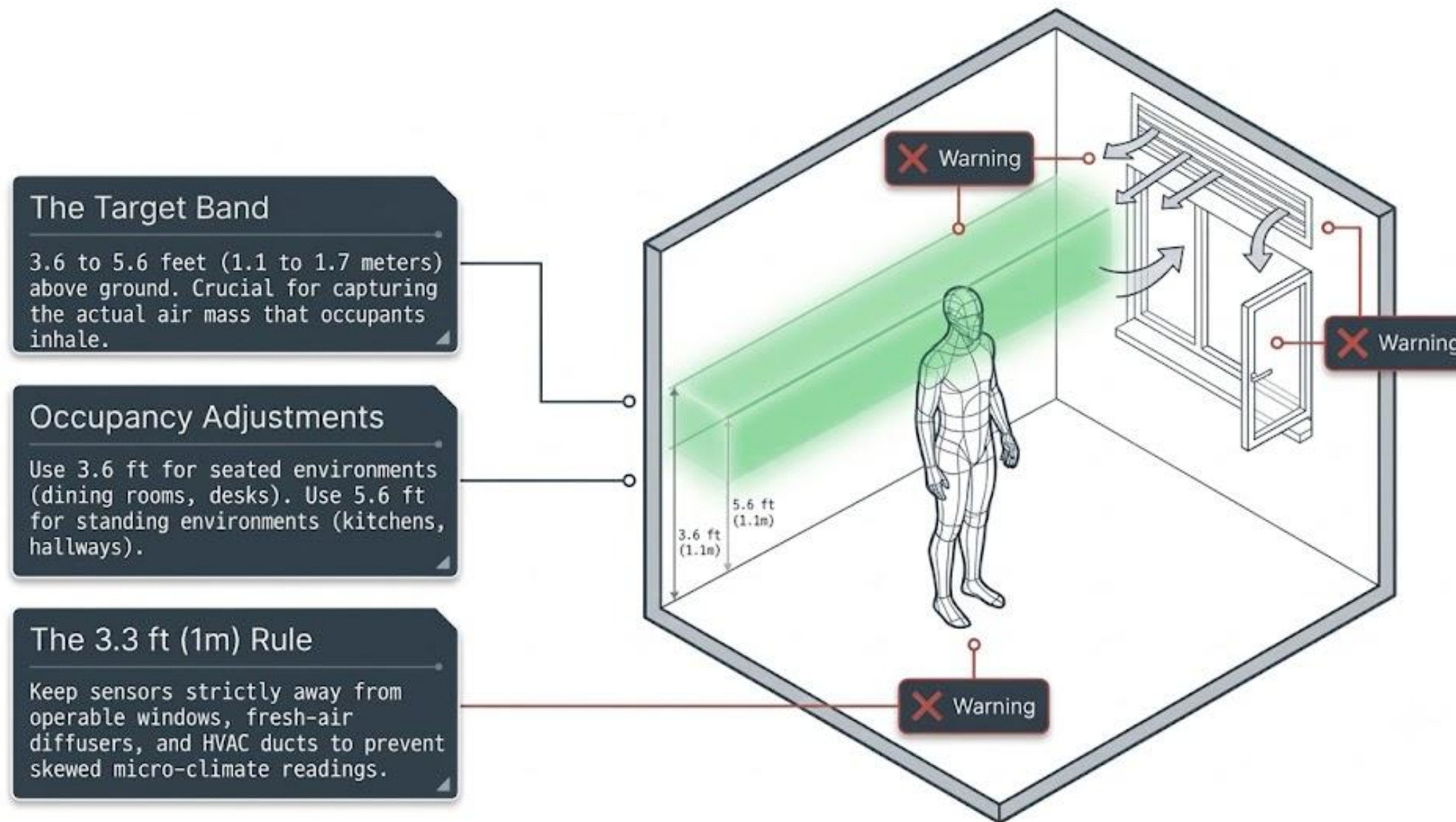
Outdoor engineering challenges



The Pillar logistics dashboard



The optimal breathing zone (indoor)



Spatial coverage blueprints



The Baseline

1 monitor per 3,500 sq ft (325 sq meters) for open areas, aligning with WELL building standards.

Schools & Offices

1 monitor per enclosed classroom or conference room to accurately track CO2 buildup and its cognitive impact.

Restaurants

Strategic split required: standing height in kitchens for high TVOCs; seated height in dining areas for viral transmission tracking.

Outdoor protocol

Avoid sinks

Placing near trees or dense shrubs artificially lowers readings as vegetation filters particulate matter.

Avoid Hyperlocal sources

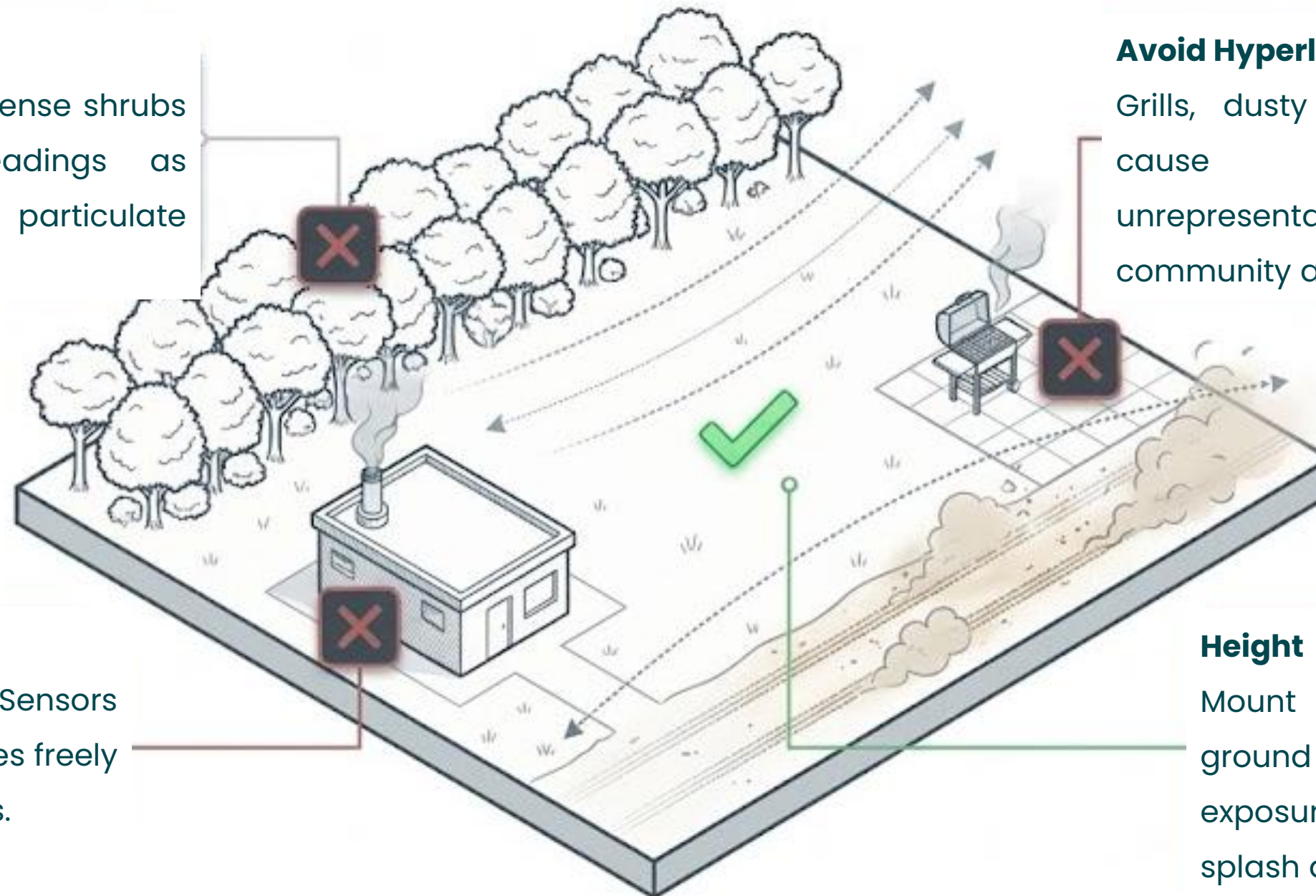
Grills, dusty roads and exhausts cause massive spikes unrepresentative of the ambient community air

Free airflow

Structure block wind. Sensors must sit where air moves freely from multiple directions.

Height

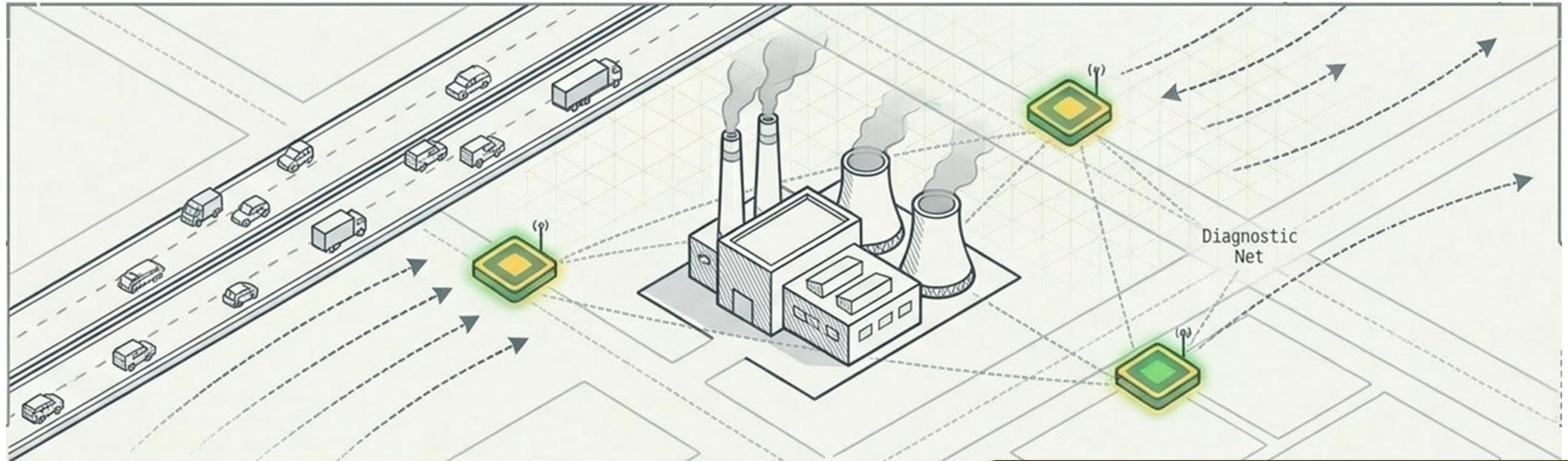
Mount 1 m to 2 m above ground to measure human exposure and prevent water splash damage.



Indoors vs. Outdoors

Indoor Environments	Dimension	Outdoor Environments
Toasters, off-gassing furniture VOCs, and human CO2 buildup.	Primary Sources	Diesel generators, busy roadways, and localized fire pits.
Air purifiers and HVAC return vents.	Primary Sinks	Tree barriers and dense vegetative walls.
Drafts from doors and fluctuating HVAC diffusers.	Micro-Climate Disruptors	High voltage lines and adjacent buildings blocking wind flow.
Tampering by pupils/public and complex WiFi splash pages.	Security & Access constraints	Theft, extreme weather exposure, and cellular dead zones.

Topology and source isolation



Upwind vs. Downwind

Place sensors both before and after a suspected pollutant source. The mathematical differential isolates and proves the exact contribution of the source.

Spatial Coverage

Disperse nodes systematically across varying local topographies and demographics to capture gradient changes in the air mass.

Community Input

Leverage local knowledge (e.g., Brightline Defense methodologies) to place nodes exactly where vulnerable populations actually spend their time.

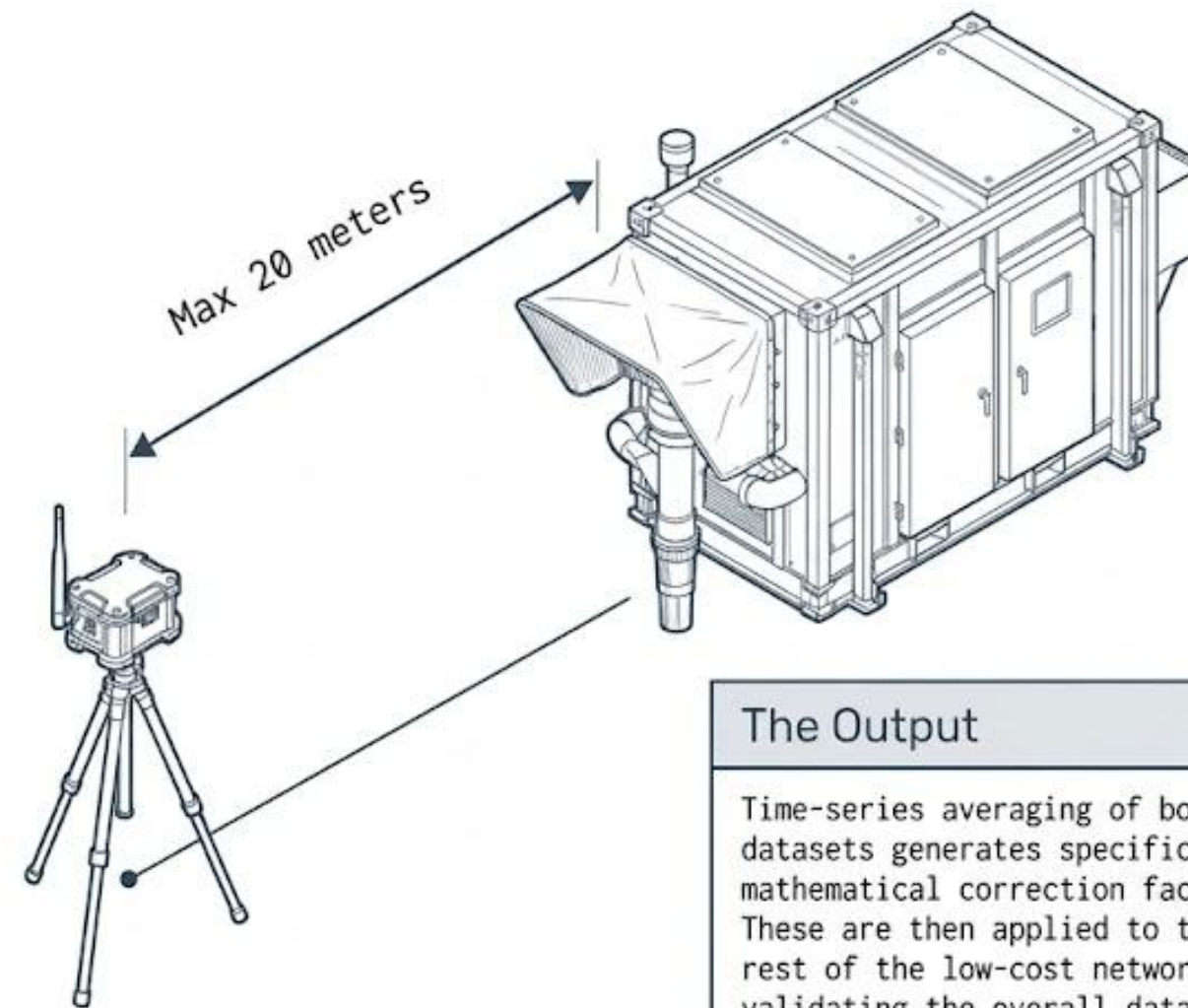
Calibration and collocation positioning

The Objective

Operating a low-cost sensor directly alongside a regulatory grade reference monitor under real-world conditions to rigorously evaluate accuracy.

The Requirement

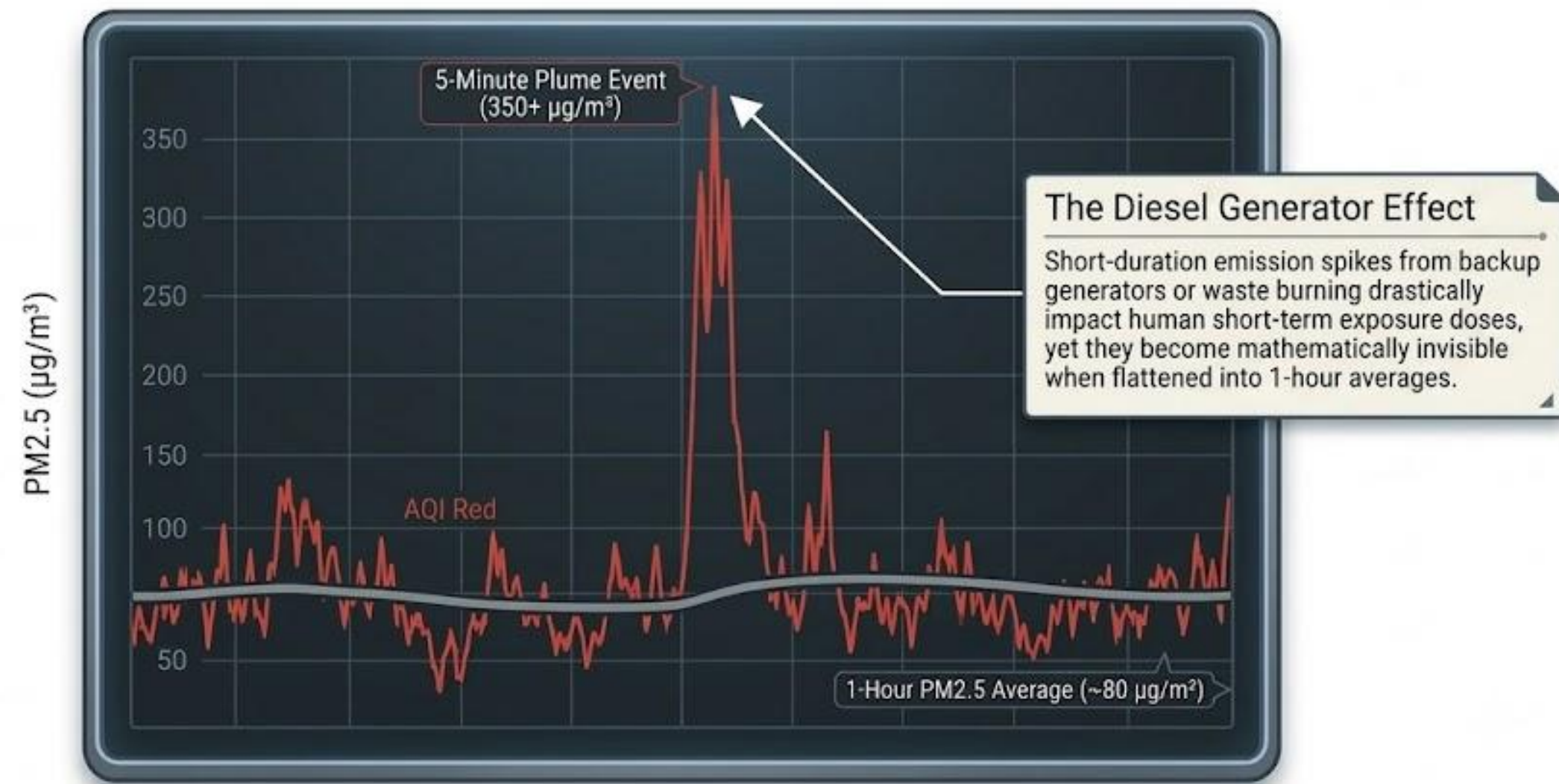
Devices must be placed within exactly 20 meters of each other outdoors. This ensures both instruments are measuring the exact same air mass without micro-climate variations.



The Output

Time-series averaging of both datasets generates specific mathematical correction factors. These are then applied to the rest of the low-cost network, validating the overall data.

Catching the plume event



The intelligent sensor network





Consortium



Ascalia



diadikasia
BRIGHT GROUP
MEMBER



smartsense

Megasense



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Thank You!



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